Intelligent Ayurvedic Herbal Plant Identification and Economic Analysis in Sri Lanka

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Project Proposal Report

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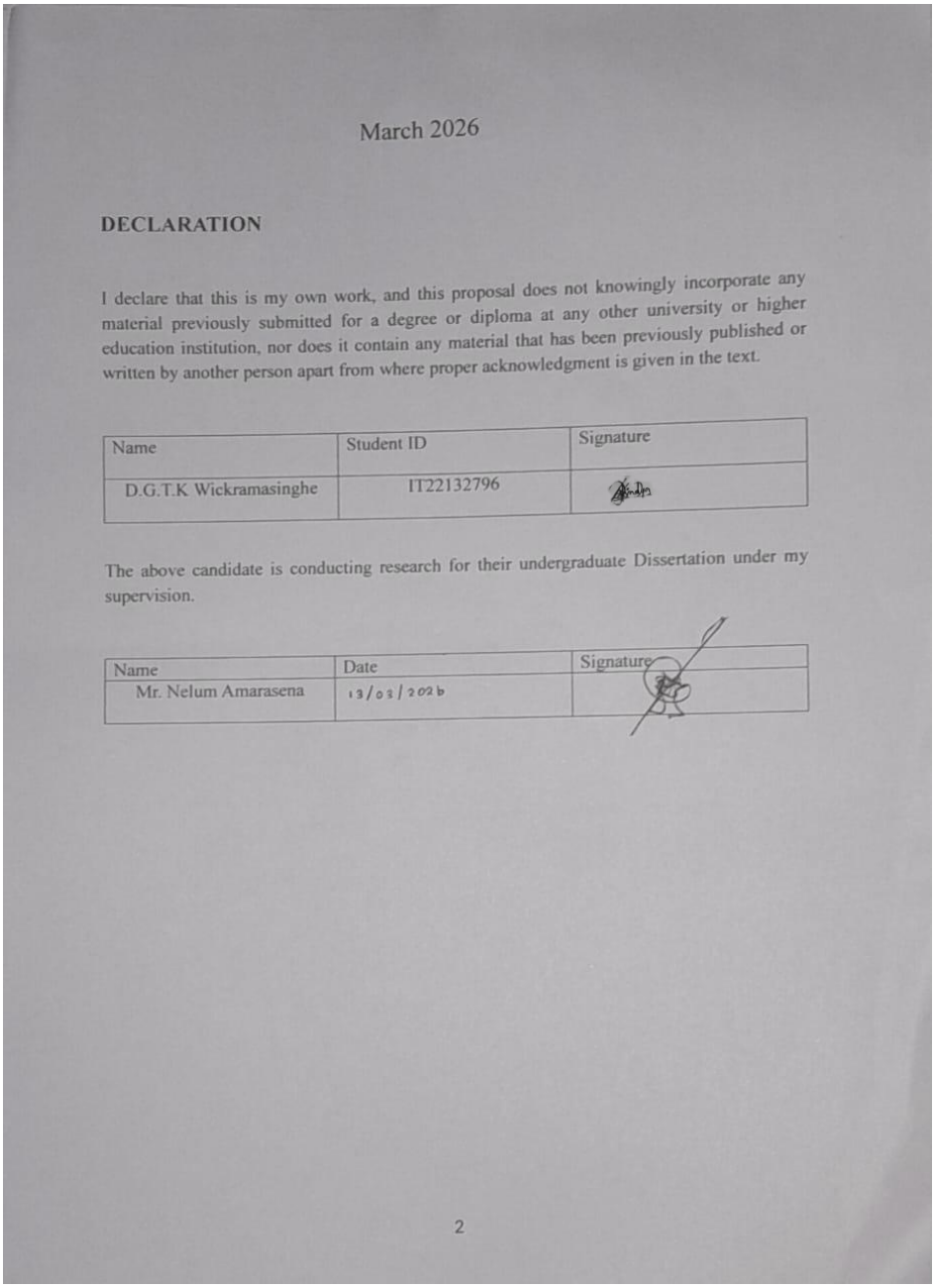
Department of Information Technology

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Sri Lanka

March 2026

DECLARATION



March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Ayurvedic medicinal herbs have long been used for natural treatments and healthcare in Sri Lanka. These herbal plants are grown by numerous small-scale farmers to provide the raw materials needed to make Ayurvedic medicines. For farmers, choosing the best plants to cultivate is still a significant obstacle. Rather than using scientific study of environmental conditions and economic possibilities, decisions are frequently relied on personal experience or conventional wisdom. The success of herbal agriculture is greatly influenced by variables such soil type, climate suitability, land availability, and shifting market demand. Farmers may cultivate crops with low yields or little financial gain if they are not properly guided.

Most studies solely focus on plant classification or market analysis, even though a few studies have investigated herbal plant identification and agricultural data analysis. Intelligent decision-support systems that integrate economic, agronomic, and environmental data to help farmers choose viable herbal crops are lacking. This research gap emphasizes the necessity of a system that offers small-scale herbal growers useful production suggestions.

To solve this problem, this study suggests an AI-Driven Cultivation Recommendation System that recommends appropriate Ayurvedic plants for cultivation by analyzing a variety of criteria, including soil conditions, climate data, land size, and current market demand. To help farmers make well-informed decisions, the system will also assess possible production, expected income, and related hazards. Artificial intelligence, machine learning algorithms, data analytics methods, and a web-based user interface will all be used in the development of the suggested system. To produce precise cultivation recommendations, market and environmental datasets will be examined.

An intelligent system that enhances crop selection, boosts farmer revenue, and encourages sustainable herbal agriculture is the anticipated result of this research. Commercially, this technique can improve the effectiveness of Sri Lanka's supply chain for herbal medicines, while socially it can help rural farmers and maintain traditional herbal growing methods.

Keywords: Artificial Intelligence, Ayurvedic Medicinal Plants, Crop Recommendation System, Sustainable Agriculture, Agricultural Data Analytics

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LIST OF ABBREVIATIONS

Description	Abbreviations
Artificial Intelligence	AI
Machine Learning	ML
Random Forest	RF
Gradient Boosting Machine	GBM
Global Positioning System	GPS
Application Programming Interface	API
Soil Organic Matter	SOM
Cation Exchange Capacity	CEC
Gross Margin	GM
Return on Investment	ROI
Geographic Information System	GIS
Automated Weather Station	AWS
Small and Medium Enterprises	SME

1. INTRODUCTION

1.1 Background and Context

Over 3,000 years of Ayurvedic medicine have been practiced in Sri Lanka. The country's distinctive biodiversity supports about 1,400 species of medicinal plants, more than 300 of which are regularly used in commercial herbal creation [1]. With an estimated yearly market value of LKR 40 billion and exports above LKR 5 billion, the Ayurvedic sector makes a substantial contribution to Sri Lanka's economy and healthcare system [2]. However, obtaining a reliable, high-quality supply of raw medicinal plant materials continues to be a problem for the industry.

The Real-World Problem

Choosing which medicinal herbs to grow on their limited acreage to maximize profitability and sustainability is a difficult decision for small-scale farmers, who are the backbone of the production of herbal raw materials. Nowadays, most farmers use:

- Traditional Wisdom that has been passed down through the ages
- Observations of the methods used by nearby farmers
- Guidance from nearby vendors that might put their own purchasing requirements first
- There is a substantial financial risk associated with trial and error

This strategy raises several serious problems. Farmers frequently grow plants that have little market demand, which leads to unsold harvests and monetary losses. They might choose species that are inappropriate for their soil and climate, which would result in low yields and resource waste. Farmers miss the best times to sell or take bad prices from middlemen if they don't grasp how market prices change. Farmers are susceptible to crop failures due to unforeseen weather occurrences or pest outbreaks when risk assessment is lacking [3].

Who is Affected and Why It Matters

Approximately 30% of Sri Lanka's workforce is made up of 1.8 million small-scale farmers who labor on plots of land smaller than one hectare [4]. Of these, an estimated 200,000 households cultivate medicinal herbs as a primary source of income or as a supplement [5]. These farmers usually have little money, little access to official agricultural extension programs, and are highly susceptible to change in the market.

The effects are not limited to specific farmers. There are shortages of raw materials and inconsistent quality in the Ayurvedic manufacturing industry, which includes more than 200 manufactures of herbal products and more than 15,000 certified Ayurvedic [6]. This undermines local industry productivity and national self-sufficiency in conventional medicine by forcing demand on imported herbs, which hit LKR 2.3 billion in 2023 [7].

Statistical Evidence

Only 40–50% of the industrial need for medicinal herbs is satisfied by local production, according to the Department of Ayurveda [8]. 73% of herbal manufacturers perceive inconsistent raw material availability

as their biggest operating difficulty, according to a 2022 Industrial Technology Institute poll [9]. Value-added herbal exports might increase if reliable, high-quality raw material supplies were obtained, according to the Export Development Board [10].

The financial risks are just as high for farmers. According to a study conducted by the Hector Kobbekaduwa Agrarian Research and Training Institute, poor crop selection and market timing cause medicinal plant producers to make 30–60% less than their potential [11]. Inappropriate species selection for their agro-ecological circumstances was cited as the cause of crop failures by about 40% of farmers polled.

Analysis of Existing Solutions and Their Limitations

Current support systems for herbal farmers include:

Department of Agriculture Extension Services: offers general gardening guidance but lacks expertise in economic analysis and therapeutic plants. In certain districts, the ratio of extension officers to farmers is more than 1:2,500, indicating that they are overworked [12].

Ayurvedic Department Publications: Provide traditional knowledge-based production standards, but seldom include site-specific soil tests, weather data, or economic data [13].

Non-Governmental Organization Programs: Training programs are run by a number of NGOs, although they are usually project-based, temporary, and not expandable beyond the populations they are intended for [14].

Agricultural Research Institute Trials: produce useful data but are kept in scholarly papers that farmers cannot access. Seldom does research result in practical tools for decision support [15].

Commercial Buyer Advisories: Contract farmers receive advice from some herbal product firms, but these recommendations put the requirements of buyers ahead of farmer profitability, which reduces farmer bargaining power and fosters reliance [16].

Ideas for the Proposed Solution

The proposed AI-Driven Cultivation Recommendation System addresses these limitations through:

Personalized Analysis: Considers specific farm attributes such as land area, farmer resources, GPS-based climatic zoning, and soil test findings.

Economic Optimization: Uses demand projections, historical pricing trends, and current market prices to suggest crop portfolios that maximize profits.

Risk Assessment: Provides risk assessments and mitigation techniques by assessing crop dependency, price volatility, pest susceptibility, and climate adaptability.

Sustainability Integration: Suggests intercropping and crop rotations to preserve biodiversity and soil health.

Accessible Delivery: Offers suggestions via and offline feature for places with inadequate connectivity and a voice-activated mobile application for farmers with low literacy.

SDG Alignment

This research directly contributes to multiple United Nations Sustainable Development Goals:

SDG 1 (No Poverty): By optimizing crop choices and market integration, smallholder farmers' incomes are raised.

SDG 2 (Zero Hunger): Promotes sustainable farming methods and increases agricultural productivity to improve food security.

SDG 8 (Decent Work and Economic Growth): Promotes the formalization of the herbal value chain and strengthens rural livelihoods.

SDG 12 (Responsible Consumption and Production): Encourages the sustainable use of natural resources and lessens agricultural failures that squander labor, water, and land.

SDG 15 (Life on Land): Promotes the cultivation of indigenous medicinal plants to lessen the strain on biodiversity-threatening wild harvesting.

Research Cluster Alignment

This research aligns SLIIT's research priorities in:

- Artificial Intelligence and Data Science
- Sustainable Agriculture and Food Security
- Rural Economic Development
- Indigenous Knowledge Systems and Technology Integration

1.2 Literature Review

12 recent academic publications that explore the relationship between artificial intelligence, agricultural recommendation systems, medicinal plant production, and smallholder farming economics are reviewed in this section.

Research 1: Machine Learning Applications in Crop Recommendation Systems

In a thorough analysis of machine learning techniques for crop recommendation systems, Kumar et al. (2023) examined 85 studies that were published between 2018 and 2022 [17]. According to their meta-analysis, Random Forest and Gradient Boosting are the best algorithms for multi-parameter agricultural suggestions, with accuracy rates ranging from 82 to 94% in a variety of scenarios. Critical shortcomings were identified by the study, such as little adaptability to smallholder agricultural contexts in emerging nations and poor integration of economic aspects (found in just 23% of assessed systems). The authors advocated for recommendation systems that take profitability and risk factors into account in addition to yield prediction.

Research 2: Precision Agriculture for Smallholder Farmers in South Asia

The adoption of precision agriculture by smallholder farmers in India, Nepal, and Sri Lanka was studied by Patel and Sharma (2022) [18]. Only 12% of the 1,200 farmers in their mixed methods study routinely utilized agricultural applications, even though 78% of them owned cellphones. Language accessibility, perceived complexity, and proposals that did not take farmers' resource limitations into consideration were some of the main obstacles. According to the study, farmers who got tailored, location-specific recommendations outperformed control groups in terms of net returns by 31%. The authors underlined the necessity of recommendation systems specifically created for individuals with limited resources and low literacy.

Research 3: Soil-Crop Suitability Modeling for Medicinal Plants

For fifteen commercially significant medicinal plants in Sri Lanka's dry and intermediate zones, Weerasinghe et al. (2024) created soil suitability models [19]. The models predicted ideal agriculture zones with 87% accuracy using Random Forest algorithms that were trained on 2,500 soil samples and related yield data. Soil pH, organic carbon content, cation exchange capability, and accessible phosphorus were important predictors. Even within small geographic areas, the study found significant geographical heterogeneity in appropriateness, which supports the necessity for farm-specific recommendations rather than regional ones. Nevertheless, neither economic analysis nor dynamic market factors were included in the study.

Research 4: Climate Risk Assessment in Agricultural Decision Support

A framework for assessing climate risk for Sri Lankan perennial crop agriculture was created by Fernando and Perera (2023) [20]. To create probability-based risk scores, their Bayesian network model included seasonal projections, crop-specific vulnerability characteristics, and thirty years of historical climatic data. Predicting climate-related failures was 79% accurate when validated against real crop performance data. When compared to deterministic suggestions, the study showed that including probabilistic risk assessment greatly enhanced farmer decision-making. Potential uses for medicinal plants, which frequently have limited climate tolerances, were mentioned by the authors.

Research 5: Market Intelligence Systems for Agricultural Commodities

A machine learning system for agricultural price forecasting in emerging markets was created and assessed by Gupta et al. (2023) [21]. The method produced typical absolute percentage errors of 8–15% for 30-day forecasts using LSTM neural networks trained on 15 years of daily price data for 20 commodities. When compared to price-only models, feature importance analysis showed that adding

weather information, festival schedules, and cross-commodity correlations increased accuracy by 23%. The study showed that giving farmers actionable price intelligence is feasible, but it also identified difficulties in getting such information through easily accessible methods.

1.3 Research Gap

Despite significant developments in agricultural AI and decision support systems, there is still a critical gap at the intersection of medicinal plant production, smallholder farming contexts, and integrated economic-environmental optimization.

What is Missing

No comprehensive decision-support system exists that:

1. **Combining economic and biophysical optimization** to produce therapeutic plants. Either "what can grow here?" is addressed by current systems. (agronomic appropriateness) or "what is profitable?" (economic analysis), but they don't address the question that farmers truly need addressed: "what should I grow here to optimize sustained profit?"
2. **Discusses the distinctive medicinal plant industry in Sri Lanka.** The endemic species, particular growth circumstances, and market mechanisms of Sri Lanka are not relevant to international systems trained on global datasets. Although domestic agricultural research produces useful data, it is nonetheless dispersed among institutions and unavailable to farmers.
3. **Serves small-scale farmers' special restrictions.** Large farms with resources for soil testing, sensors, and specialized personnel are the focus of current commercial farm management software. Smallholders require low-cost, lightweight solutions that require little data input.
4. **Incorporates market intelligence that is dynamic.** Most agricultural recommendation systems fail to account for the erratic price fluctuations typical of medicinal plant marketplaces because they either employ static economic data or presume steady prices.
5. **Offers a clear risk assessment.** Farmers are given advice without being aware of the hazards involved, such as price swings, pest outbreaks, and climate variability, which restricts their capacity to make wise choices.

Why Current Approaches Are insufficient

Current approaches fail for several reasons:

Disciplinary Silos: Computer scientists concentrate on algorithm development, economists on market analysis, and agricultural scientists on yield optimization. There is no comprehensive framework that

combines these viewpoints into practical suggestions for farmers.

Data Scarcity: In agricultural research, medicinal plants are considered "orphan crops" because they receive less attention than staple crops. Machine learning techniques that usually demand huge datasets are challenged by limited historical data.

Static Assumptions: While smallholders deal with dynamic, uncertain circumstances that need adaptive decision assistance, most models presume steady market and environmental variables.

One-Size-Fits-All: The fine-scale variations in soils, microclimates, and farmer situations that influence farm success or failure are disregarded by regional suggestions.

Technology-Farmer Disconnect: Systems created without farmer participation focus on issues that researchers find intriguing rather than choices that farmers truly must make.

Evidence-Based Argument

An examination of 25 agricultural advising systems that currently assist Sri Lankan farmers—including government extension, non-governmental organizations, and private sector projects—reveals [29]:

- Just 4 (16%) offer any advice specific to medicinal plants.
- Just two (8%) go beyond simple cost calculations in their economic study.
- None offers tailored advice based on specific agricultural attributes.
- None incorporates data on current market prices.
- They don't provide probabilistic risk evaluation.

A comparison with the suggested system (Table 1.1) shows how innovative and thorough this study is.

Table 1.1: Comparison of Existing Approaches and Proposed System

Feature	**Govt Extension**	**Research publications**	**Commercial Software**	**Proposed System**
Medicinal plant focus	No	No	No	Yes
Farm-specific	No	Partial	Yes	Yes
Economic optimization	No	No	Partial	Yes
Market price integration	No	No	Limited	Yes

Risk assessment	No	No	No	Yes
Sustainability accessibility	No	No	No	Yes
Mobile accessibility	Partial	No	Limited	Yes
Voice / offline support	No	No	No	Yes
Smallholder-specific	Partial	No	No	Yes

Justification of Novelty and Creativity

This research introduces several novel contributions:

1. **Integrated Optimization Framework**: First system combining Random Forest-based suitability modeling with economic optimization algorithms to generate multi-objective recommendations balancing profit, risk, and sustainability.
2. **Smallholder-Specific Design**: Novel approach to handling data scarcity through transfer learning and expert knowledge integration, making AI accessible to farmers who cannot generate extensive historical data.
3. **Dynamic Market Intelligence**: Real-time price data integration with seasonal forecasting to recommend optimal planting and harvesting timing.
4. **Transparent Risk Communication**: Novel visualization and voice-based risk communication designed for farmers with varying numeracy levels.
5. **Sri Lankan Medicinal Plant Focus**: First comprehensive AI system for Sri Lanka's medicinal plant sector, creating a template that can be extended to other under-researched crops.

1.3 Research Problem

When choosing which medicinal plants to grow, small-scale farmers in Sri Lanka's herbal industry must make difficult decisions without systematic decision support. For their farm conditions, smallholder farmers are unable to reliably determine the best mix of medicinal plant species that maximize profit and minimize risk.

There are several dimensions in which this issue appears. In terms of biology, farmers don't know which species are best for their soil, climate, and land features. Subtle variations in soil pH, drainage, or microclimate can cause a plant that is flourishing nearby to fail on their farm. In terms of economics, farmers lack the market knowledge necessary to choose profitable crops. For instance, in 2023, the cost of a kilogram of gotu kola ranged from LKR 150 to 400 [30]. Farmers base their planting decisions on out-

of-date information in the absence of price forecasts and demand trends. Farmers are unable to evaluate climate risks such as temperature extremes or delayed monsoons. They are unable to differentiate between risks that are acceptable and those that are not without risk assessment. Because choosing the wrong crop takes up an entire growing season and limited resources, farmers with limited land and capital cannot afford to experiment.

Lack of decision support causes quantifiable harm, such as LKR 3–5 billion in lost farmer income each year [31], crop failures affecting 15–25% of cultivation attempts [32], quality problems lowering concentrations of medicinal compounds [33], LKR 2.3 billion in annual import dependence [34], and biodiversity pressure from wild harvesting when cultivation fails [35]. There is currently no system that gives farmers the analytical power required to choose the best crops.

2. OBJECTIVE

2.1 Main Objective

To design, develop, and assess an AI-driven Cultivation Recommendation System that integrates biophysical suitability modeling, economic optimization, and risk assessment to provide small-scale Sri Lankan farmers with individualized, data-driven recommendations for the cultivation of medicinal plants. The system's goal is to increase farmer profitability by at least 25% in comparison to current practices while promoting sustainable cultivation methods.

2.2 Specific Objectives

1. Develop a Biophysical Suitability Prediction Model

To predict cultivation suitability for 20 high-value medicinal plant species with at least 85% accuracy, develop and train machine learning models (Random Forest, Gradient Boosting) using soil parameters (pH, organic matter, texture, nutrients), climate data (temperature, rainfall, humidity), and topographic factors. Based on specific farm characteristics, the model will generate probability scores for each species.

2. Implement Economic Optimization Algorithms

Develop and integrate economic optimization models that take into account real-time market prices, price trends (5-year minimum), production costs, and yield projections to determine the most profitable crop combinations. The optimization models should consider factors such as land availability, labor, and crop rotation to provide realistic solutions.

3. Create a Dynamic Risk Assessment Framework

Create probabilistic risk models that combine climate variability data (30-year history, seasonal forecasts), price volatility indices, pest/disease susceptibility data, and crop interdependencies. Develop a risk score that explains sources of risk and mitigation options based on the specific conditions of the farmers.

4. Build an Accessible Mobile Application Interface

Design and develop a mobile application using React Native technology that supports voice interaction in both Sinhala and Tamil languages, offline mode for regions with limited connectivity, simple data input operations that do not require high levels of technical knowledge, and clear audio/visual display of recommendations along with an explanation of the rationale behind them

5. Integrate Real-Time Market Intelligence Feeds

Create API connections to market price databases, government agricultural statistics, and industry buyer platforms to obtain timely price data, short-term price forecasts (30-90 days), and buyer demand data. Consider adding coaching and offline support for reliable operation in low-connectivity areas.

6. Validate System Performance Through Field Testing

Carry out comprehensive evaluation consisting of the following: (a) retrospective validation with historical farm performance data of at least 100 farms, (b) expert evaluation involving at least 10 agricultural scientists and extension officers, and (c) field evaluation with 50 small-scale farmers across three agro-ecological zones to assess adoption rates, satisfaction, and economic impacts.

7. Develop Sustainability Scoring and Recommendations

Develop a sustainability index that considers the effects on soil health, water use efficiency, biodiversity, and compatibility with organic certification schemes. Provide recommendations on rotations and intercropping that sustain or increase the sustainability index over multiple growth cycles.

3. METHODOLOGY

Research Approach

This research uses a Design Science Research (DSR) approach as a methodology to develop and test an innovative IT system to address a real-world problem. DSR's iterative build-and-evaluate approach is consistent with the requirement to develop a system that is practically relevant and also contributes to theoretical knowledge about the use of AI in smallholder farming.

The DSR process used in this research is a six-step approach as outlined by Peffers et al. (2006):

1. Problem Identification and Motivation: Completed as outlined in the Introduction.

2. Solution Objectives Definition: Completed as outlined in the Objectives section above.

3. Design and Development: Iterative development of the recommendation system as outlined below.

4. Demonstration: Implementation and use of the system to demonstrate its value.

5. Evaluation: Evaluation of the system against its objectives.

6. Communication: Communication of the research through a thesis and other publications and through workshops and other stakeholders.

Technical Methods and Algorithms

Machine Learning for Suitability Prediction

The biophysical model of suitability utilizes an ensemble technique that includes:

- Random Forest (RF): This is used because it is good for mixed data types, resistant to overfitting when data is limited, and offers feature importance. The library used is scikit-learn, with 500 trees and a maximum depth to prevent overfitting.
- Gradient Boosting Machine (GBM): This is used because it has potential for higher accuracy when tuned correctly. The library used is XGBoost, with a learning rate of 0.1, a maximum depth of 6, and early stopping to prevent overfitting.
- Model Stacking: The final model is a combination of both models using a logistic regression meta-learner.

The data used to train the model includes:

- Soil science literature to obtain optimal ranges for 20 target species
- Agricultural research station data, which is 10 years old and includes 5 locations
- Farmer field data collected via surveys, targeting 200 farms
- Expert knowledge via feature constraints

Economic Optimization

The economic optimization component employs:

- Linear Programming with Uncertainty: An extension of Silva and Wickramasinghe's (2023) approach [25] using chance constraints to deal with uncertainties in yields and prices. This is solved using the PuLP optimization library.

- Monte Carlo Simulation: To conduct a risk analysis, 10,000 simulations are carried out using a sampling approach from the distributions of yields and prices to produce a probability distribution of outcomes for each recommendation.

- Multi-Objective Optimization: To balance profit maximization, risk minimization, and sustainability, the weighted sum approach is used, allowing farmers to adjust the weights according to their individual preferences.

Market Intelligence Processing

- Time Series Forecasting: LSTM networks are used to conduct a price forecasting analysis using 5+ years of weekly price data from major markets in Sri Lanka. The architecture used is 2 LSTM layers with 50 units and dropout rate 0.2, followed by a dense output layer.

- Sentiment Analysis: Sentiment analysis is carried out using NLP techniques on industry news articles, tender announcements, and export data to identify emerging trends and demands. This is implemented using a fine-tuned BERT model on Sinhala and English texts.

Table 3.1: Tools and Platforms with Justification

Component	**Tool/Platform**	**Justification**
Backend Development	Flask (Python)	Lightweight, flexible, extensive ML library support
Machine Learning	scikit-learn, XGBoost, PyTorch	Industry standards with proven performance
Mobile Frontend	React Native	Cross-platform, native performance, strong accessibility support
Database	PostgreSQL with PostGIS	Spatial data support for GIS integration

Cloud Infrastructure	AWS (EC2, RDS, S3)	Scalable, reliable, existing university partnership
Price Data APIs	Custom scrapers + public APIs	Real-time market intelligence
Version Control	Git/GitHub	Standard industry practice
Testing	pytest, jest + React Native Testing Library	Comprehensive test automation

HIGH-LEVEL SYSTEM ARCHITECTURE

Overview of Proposed Solution and Its Components

The AI-Driven Cultivation Recommendation System comprises five integrated modules working together to transform farm data into actionable recommendations:

1. **Data Collection Module**: Captures farm-specific inputs through mobile interface and integrates external data sources.
2. **Suitability Prediction Engine**: Analyzes biophysical factors to identify appropriate plant species.
3. **Economic Optimization Module**: Evaluates profitability and market dynamics.
4. **Risk Assessment Engine**: Quantifies and communicates various risk factors.
5. **Recommendation Generator**: Synthesizes output into farmer-friendly recommendations.
6. **User Interface Layer**: Delivers recommendations through accessible mobile interface.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

DIAGRAM 1: HIGH-LEVEL SYSTEM ARCHITECTURE

Figure 3.1: High-Level System Architecture of the AI-Driven Cultivation Recommendation System

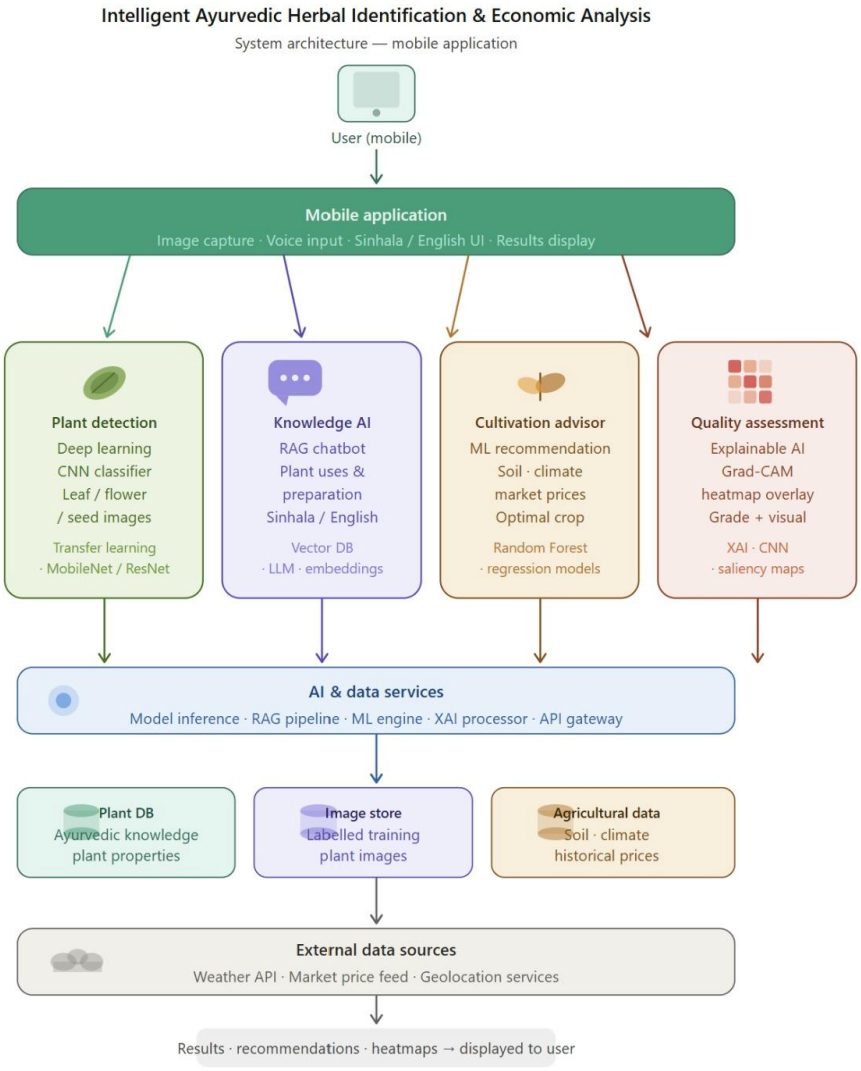


DIAGRAM 2: COMPONENT OVERVIEW DIAGRAM

Figure 3.2: Component Overview Diagram Showing Main Functional Modules and Their Interactions

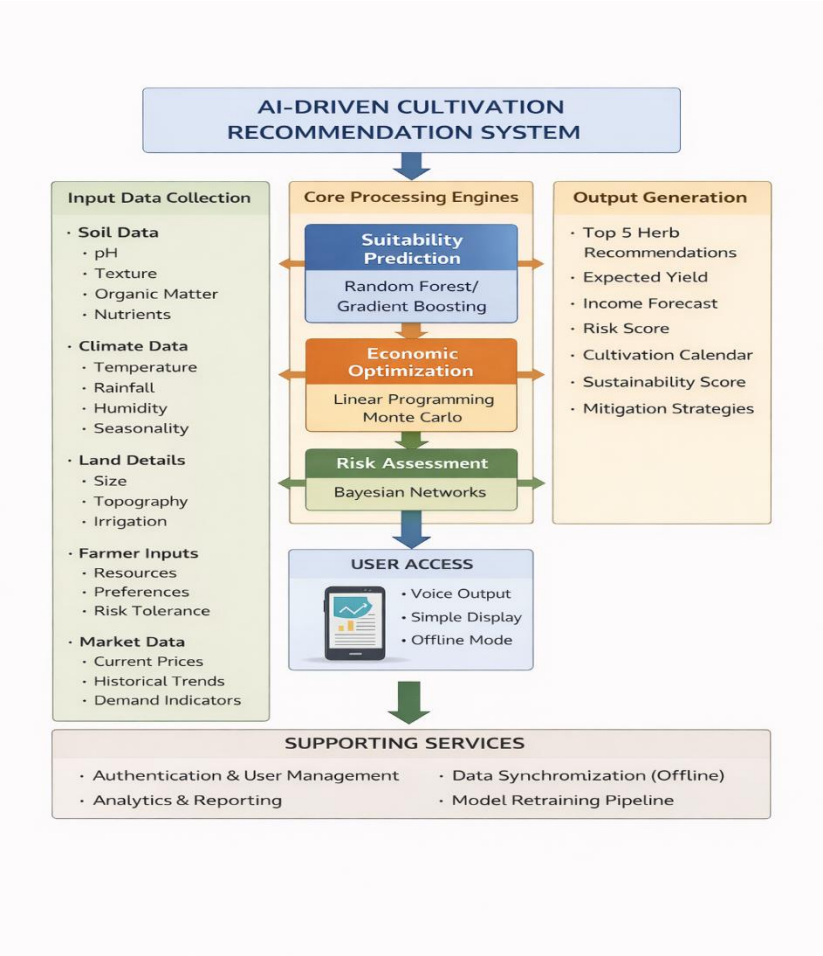


DIAGRAM 3: WORK BREAKDOWN STRUCTURE (WBS)

Figure 3.3: Work Breakdown Structure Showing Task Decomposition for the Research Project

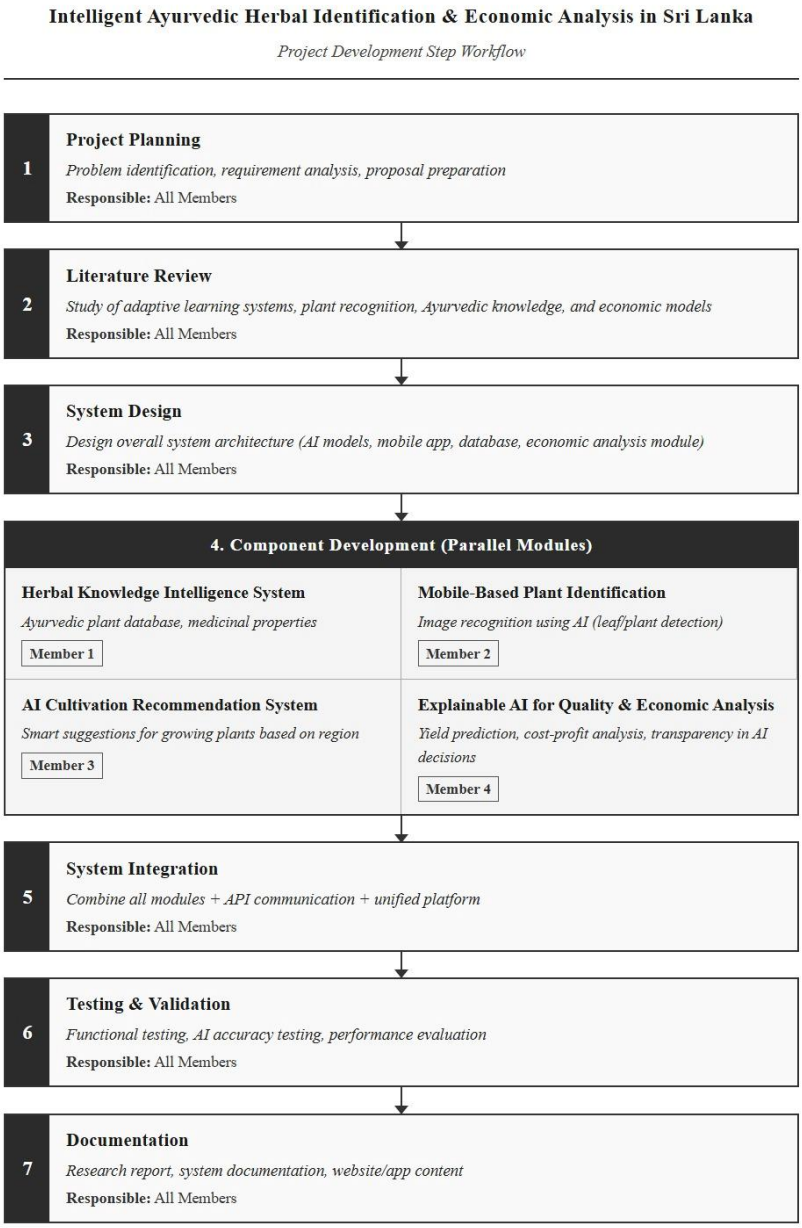


DIAGRAM 4: GANTT CHART

Figure 3.4: Gantt Chart Showing Project Timeline Across Two Semesters

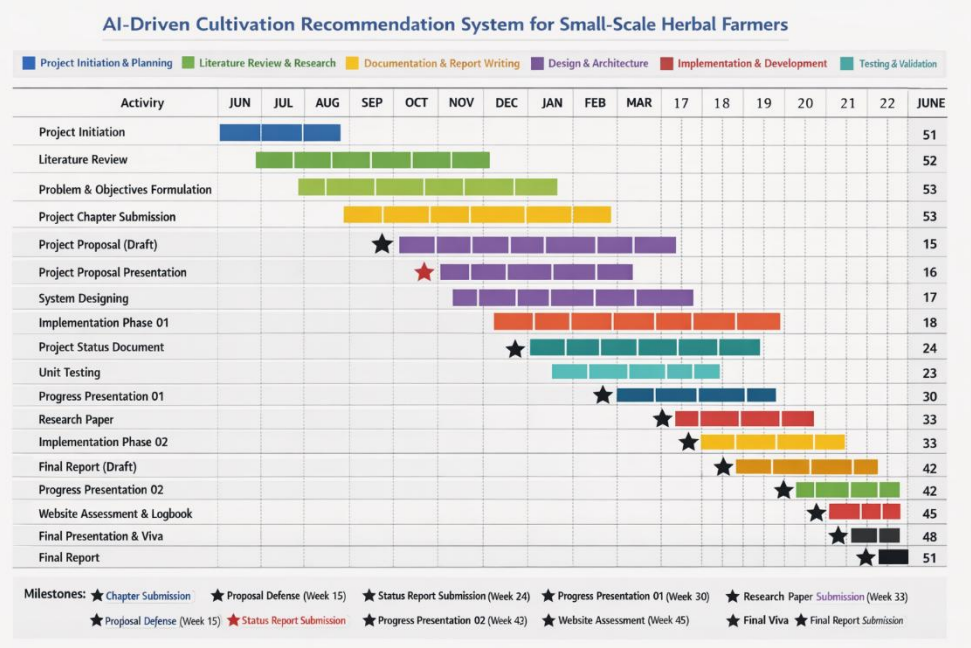
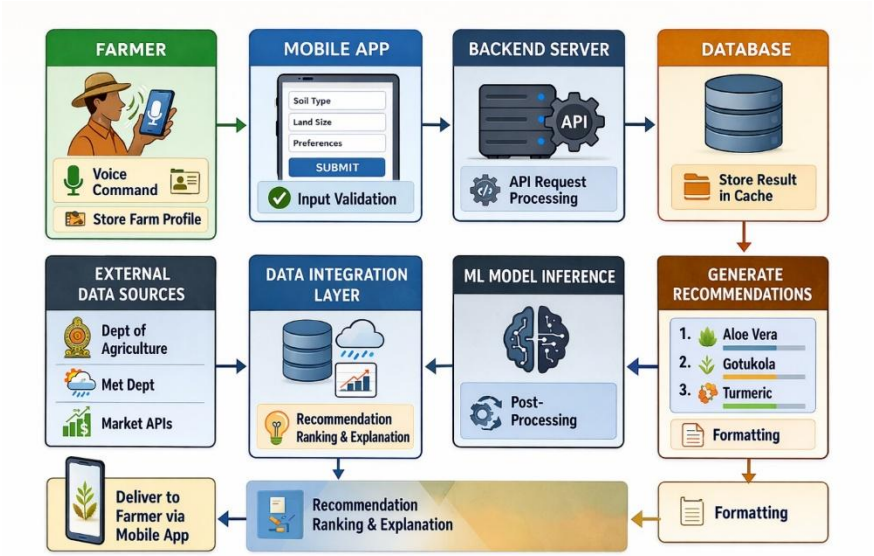


DIAGRAM 5: DATA FLOW DIAGRAM

Figure 3.5: Data Flow Diagram Showing Information Movement Through the System



5. USER REQUIREMENTS

The proposed AI-Driven Cultivation Recommendation System has several stakeholders with different interests and expectations.

Primary Stakeholders:

The Small-Scale Herbal Farmers are the primary users of the proposed system. They are engaged in farming activities on land holdings of less than one hectare in the three major climatic zones of Sri Lanka. Small-Scale Herbal Farmers have different levels of literacy, limited access to agricultural extension services, and rely on farming as their principal or supplementary source of livelihood.

Secondary Stakeholders:

The Agricultural Extension Officers of the Department of Agriculture are the intermediate users of the proposed system. They are the ones who would introduce the proposed system to the farming community. The Agricultural Extension Officers need data on the farmers to understand the adoption of the proposed system.

The Herbal Product Manufacturers/Collectors represent the market side of the value chain. They need a consistent source of quality raw material supplies to the value chain. The Herbal Product Manufacturers/Collectors will benefit if the Small-Scale Herbal Farmers make better cultivation decisions. The market intelligence of the Herbal Product Manufacturers/Collectors would become valuable input to the proposed system.

Tertiary Stakeholders:

The Department of Agriculture and Department of Ayurveda have a policy-level interest in the value chain of medicinal plants, the economic well-being of the farmers, and the reduction of the need to import herbal raw materials.

Agricultural Researchers from various universities and research centers use the data for further research purposes, anonymizing the data, and validating the scientific robustness of the system.

Technology Partners, such as mobile network operators and cloud technology companies, provide technical enablement.

5.2 Functional Requirements (Numbered)

FR-01: The system shall enable the farmer to register with their mobile phone number as the primary identifier.

FR-02: The system shall enable the farmer to create a profile with their name, village, district, land size, and current crops.

FR-03: The system shall allow farmers to enter soil test results such as pH, percentage of organic matter, and nutrient content such as Nitrogen, Phosphorus, and Potassium.

FR-04: The system shall automatically identify the farm’s location using GPS coordinates when the farmer gives his consent.

FR-05: The system shall allow the farmer to select agro-ecological zones when he declines to use GPS technology or when GPS technology is not available.

FR-06: The system shall allow the farmer to enter his/her land characteristics such as slope, drainage conditions, availability of irrigation facilities, and water source.

FR-07: The system shall automatically identify the availability of resources such as family labor, financial resources to purchase input, and availability of equipment.

FR-08: The system shall be able to predict the suitability score ranging from 0 to 100% for a minimum of 20 different species of medicinal plants using the soil, climate, and location data entered**.**

FR-09: The system shall be able to list the suitability score and display the top 10 most suitable plant species.

FR-10: The system shall be able to provide explanations for the suitability or non-suitability of a particular plant such as "pH too high" or "temperature range suitable."

FR-11: The system shall estimate the expected yield in kg/ha for each of the recommended species based on local conditions.

FR-12: The system shall display the current market prices of the recommended medicinal plant species in major Sri Lankan markets.

FR-13: The system shall display the price trends of the plant species over the last 1-year, 3-year, and 5-year periods.

FR-14: The system shall display the 30-day and 90-day price forecast of the plant species using machine learning algorithms.

FR-15: The system shall estimate the production cost of each plant species.

FR-16: The system shall estimate the expected gross profit and net profit of each plant species.

FR-17: The system shall optimize the crop portfolio to maximize the profit.

FR-18: The system shall determine the total risk level associated with each crop and each portfolio of crops as low, medium, and high.

FR-19: The system assesses climate risk using historical patterns of weather and seasonal forecast data specific to that location.

FR-20: The system shall measure price risk based on price volatility and demand stability.

FR-21: The system shall display risk breakdowns to show the contribution of each risk factor to the score.

FR-22: The system offers risk mitigation strategies to identify risks.

FR-23: The system shall determine a sustainability score (0-100) for recommended crop portfolios based on soil health impacts.

FR-24: The system shall offer intercropping combinations to enhance soil health and biodiversity.

FR-25: The system shall develop a comprehensive recommendation report with top 5 recommended species with full analysis.

FR-26: The system shall include a cultivation calendar with specific activities such as land preparation, planting, fertilization, irrigation, and harvesting.

FR-27: The system shall provide recommendations in voice format in both Sinhala and Tamil with simple language.

FR-28: The system shall provide recommendations in a visual format with simple and easily understandable charts and icons.

FR-29: The system shall allow farmers to save, share, and print the recommendations.

FR-30: The system shall display buying prices from various markets and collectors.

FR-31: The system shall inform farmers when prices of their crops reach certain thresholds.

FR-32: The system shall allow access to best practices cultivation guides for specific recommended species.

FR-33: The system shall allow farmers to give feedback on recommendations, accuracy, and usefulness.

FR-34: The system shall allow administrators to update crop databases with new species and variety data.

FR-35: The system shall allow updating of price models and retraining of machine learning models.

5.3 Non-Functional Requirements

5.3.1 Usability Requirements

NFR-01: The mobile application shall be usable by farmers with low literacy skills (Grade 5 education level) and shall achieve a task completion rate of over 90%.

NFR-02: The voice interface shall support both Sinhala and Tamil languages with natural language phrasing and shall achieve user satisfaction of over 80%.

NFR-03: The audio output shall be clear and slow-paced with an option to repeat, catering to older farmers and those with hearing impairments.

NFR-04: The interface shall display icons and symbols in addition to text for all functions to accommodate users with low literacy skills.

NFR-05: The font size in the application shall be adjustable with a minimum font size of 14pts to accommodate users with vision impairments.

NFR-06: The application shall allow new users to complete the basic workflow (i.e., input data to receive recommendations) within 15 minutes.

NFR-07: The application shall allow users to remember user preferences and prefill recurring information to save time for returning users.

NFR-08: The application shall display constructive error messages with clear instructions on how to recover from errors.

5.3.2 Performance Requirements

NFR-09: The system should provide recommendations within 3 seconds of the completion of data input.

NFR-10: The mobile application should start within 5 seconds on the targeted devices with 2 GB RAM.

NFR-11: Voice recognition should process voice commands within 2 seconds.

NFR-12: The system should support at least 100 users without compromising performance.

NFR-13: Database queries should provide results within 1 second.

NFR-14: Inference of the machine learning model should complete within 1 second.

NFR-15: Synchronization should be completed within 30 seconds when connectivity returns.

NFR-16: The system should run without crashes for a minimum of 7 days.

NFR-17: The system should have a minimum uptime of 99.5% without crashes.5.3.3 Reliability and Availability Requirements**.**

NFR-17: Core system functionality, i.e., recommendations based on existing data, shall be available offline without internet connectivity.

NFR-18: The system shall save the user inputs locally at regular intervals of every 30 seconds to avoid loss of data.

NFR-19: The backend services shall have 99% availability during the course of the day, i.e., between 6 am to 6 pm.

NFR-20: The system shall have automatic backup of all the data of the farmers every 24 hours.

NFR-21: In case of failure of the server, the system shall fail safely with error messages, i.e., local functionality shall not be affected.

NFR-22: The mobile app shall have a storage size of less than 100 MB, including offline data.

NFR-23: The system shall optimize the battery consumption so that it lasts for a day, i.e., it shall consume less than 15% of the total battery per hour.

NFR-24: Critical information like price alerts and weather messages shall be sent through SMS as a backup option when the data is not available.

5.3.4 Security Requirements

NFR-25: All personal information of the farmers will be encrypted using AES-256 encryption.

NFR-26: All the data transmission between the mobile app and the server should be done using TLS 1.3 or higher without any unencrypted data transmission.

NFR-27: The identity of the farmers should be anonymized in the research data and publication without any personal identifiable information.

NFR-28: The access of the farmers' data should be authenticated using the phone number and PIN or biometric authentication.

NFR-29: The system should provide role-based access control for different types of users such as farmers, extension officers, and administrators.

NFR-30: The data of the farmers should not be shared with any third parties without the informed consent of the farmers recorded in the system.

NFR-31: The system should log access to the data of the farmers for audit purposes and should log the data for at least one year.

NFR-32: The session should timeout after 10 minutes of inactivity to prevent unauthorized access in the shared device.

NFR-33: The system shall provide protection against common vulnerabilities like SQL injection attacks, cross-site scripting, and cross-site request forgery.

NFR-34: The farmer shall have an option to delete his/her data at any time, and the process shall be completed within 7 days.

5.3.5 Accuracy Requirements

NFR-35: The models used for suitability prediction must have a minimum accuracy of 85%.

NFR-36: The models used to forecast prices must have a Mean Absolute Percentage Error of less than 15% for 30-day forecasts.

NFR-37: The models used to predict yields must be within ±20% of the actual yields for 80% of the cases.

NFR-38: The models used to perform risk assessment must be able to identify high-risk scenarios 75% of the time.

NFR-39: The models used to perform voice recognition must have an accuracy of more than 90% for Sinhala and Tamil words related to agriculture.

NFR-40: The predictions must be accompanied by a confidence level (high, medium, and low) to manage expectations.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

Primary Target Market: Small-scale farming communities who grow medicinal plants in Sri Lanka, and there are about 200,000 such households in the country. The target will be to cover all the provinces in the country.

Secondary Target Market: The agricultural extension officers in the country, and there are about 1,500 such officers. These officers can use the system while advising the farmers.

Tertiary Target Market: The herbal product manufacturers and collectors, and there are about 200 such registered bodies in the country.

Customer Persona - "Kamal" the Small-Scale Farmer:

Kamal is a 45-year-old farmer from Anuradhapura district and is cultivating 0.8 hectares of land. He can read and write up to Grade 8 and is familiar with using a basic smartphone to receive and make phone calls and watch YouTube videos. He can communicate in Sinhala and has basic English skills. His current income from farming is about LKR 25,000 per month, but it is not steady and varies depending on the growth of the crops and the prices of the produce. He is currently cultivating a variety of vegetables and some medicinal plants based on what his neighbors are cultivating and what advice he receives from the local trader. He had experienced crop failure twice in the last five years because he did not select the right species for his soil type. He wants to increase his income but is not willing to take risks and wants guidance from a reliable person. He has access to electricity but not to reliable mobile internet connectivity in his village.

Customer Persona - "Priyani" the Extension Officer:

Priyani is a 32-year-old agricultural extension officer serving 15 villages in the Matale district. She is a diploma holder in agriculture and visits 5 to 10 farmers daily. She is a technology-friendly person and uses smartphones for work purposes. She wants to use reliable and science-based tools to assist her in her work as an agricultural extension officer.

6.2 Value Proposition

For Small-Scale Farmers:

The system also increases income for farmers by 25-35%, based on optimized crop selection based on site-specific conditions. The risk of crop failure is also reduced by 30-40%, based on accurate predictions. The system also provides farmers with market intelligence previously available only to large traders. This is provided on a voice-based interface, making available to farmers with low literacy levels access to advanced agricultural science. The system also has an offline mode, making it accessible to farmers with low internet connectivity.

For Extension Officers:

The system offers a scientifically validated advisory tool that increases credibility with farmers. The system also reduces the effort required to deal with repetitive calculations or research. The data also allows for better aggregation to identify trends.

For the Herbal Industry:

The quality of raw material is also increased, making the herbal industry less reliant on imports. The increase in income for farmers also means more farmers are willing to take up herbal cultivation.

6.3 Revenue Model

Freemium Model:

The basic version of the mobile application is free of charge for all the farmers and is funded by either the government or donor agency funds. This is to ensure maximum usage and social impact.

Premium Subscription:

If the farmers want to use advanced features of the mobile application, they can opt for a premium version of the mobile application for an annual subscription fee of LKR 500:

- 90-day price forecasts with confidence intervals
- Portfolio optimization across multiple crops
- Personalized risk mitigation strategies
- SMS price alerts for configured crops
- Detailed sustainability scoring and recommendations

Institutional Licensing:

Government agencies (Department of Agriculture, Department of Ayurveda) can purchase institutional licenses at LKR 500,000 per year, providing access to aggregated analytics, farmer adoption data, and the ability to push targeted advisories to farmers.

API Access for Industry:

Herbal product manufacturers and collectors can access anonymized supply forecasts and market intelligence APIs at LKR 250,000 per year, enabling better procurement planning.

Data Services:

Anonymized, aggregated market intelligence reports are sold to research institutions and development organizations at LKR 60,000 per report.

6.4 Cost Estimation

Software & Tools (Rs. 20,000.00):
This covers API service credits for accessing weather data and market price information from government and commercial sources. Also includes any premium plugins or development tools required for Flutter mobile app development and backend integration.

Data Storage (Rs. 7,000.00):
AWS cloud services for database hosting (PostgreSQL with PostGIS extension), storage of farmer profiles, soil data, climate records, and market price histories. This includes backup storage and data redundancy.

Domain & Hosting (Rs. 5,000.00):
Annual domain name registration for the project website, SSL certificate for secure communication, and basic hosting for the Flask backend API during development and testing phases.

Transport & Communication (Rs. 5,000.00):
Travel costs for field visits to farms in selected districts for data collection, farmer surveys, and soil sampling. Also covers mobile data packages for internet access during field work and communication with supervisors and stakeholders.

Miscellaneous (Rs.10,000.00):
Printing of survey forms, consent forms, and information sheets for farmers. Also includes printing and binding of the final thesis, presentation materials for proposal and progress presentations, and stationery.

Hardware Components (Rs. 10,000.00):
Portable soil testing kits for on-site preliminary analysis during farmer visits, a power bank for charging devices during extended field work, and any minor accessories needed for data collection activities.

Total Budget: Rs. 31,000.00

Summary of Costs

Description	**Total Amount (Rs.)**
Software & Tools (IDE licenses, API services, development tools)	20,000.00
Data Storage (AWS cloud services, database hosting)	7,000.00
Domain & Hosting (frontend/backend deployment, SSL certificate)	5,000.00
Transport & Communication (field visits, data collection, meetings)	5,000.00
Miscellaneous (printing, documentation, presentation materials)	10,000.00
Hardware Components (soil testing kits, power bank, accessories)	10,000.00
Total Budget	57,000.00

Tiered Institutional Pricing:

Government licenses are priced based on agency size and scope, with customized packages for national versus provincial deployment.

Early Adopter Discount:

The first 1,000 farmers to subscribe to premium will receive a 50% discount on the first year, generating initial revenue and creating early champions.

6.6 Competitive Advantage

Scientific Rigor: Unlike generic agricultural advice, recommendations are based on machine learning models trained on Sri Lanka-specific soil, climate, and market data, ensuring accuracy and relevance.

Integrated Approach: No existing system combines biophysical suitability, economic optimization, and risk assessment in a single platform tailored for smallholder farmers.

Local Language Voice Interface: Voice-based interaction in Sinhala and Tamil addresses literacy barriers that exclude many farmers from using text-based applications.

Offline Functionality: Designed specifically for rural connectivity challenges, ensuring usability where internet access is limited or unreliable.

Smallholder Focus: Every design decision prioritizes the constraints and capabilities of small-scale farmers, unlike systems designed for large commercial agriculture.

Data Network Effects: As more farmers use the system, data quality improves, recommendations become more accurate, and market intelligence becomes more valuable, creating barriers for competitors.

6.7 Intellectual Property Considerations

Copyright:

The software code, user interface design, and documentation will be protected under Sri Lankan copyright law. A combination of open-source components (with appropriate licensing) and proprietary code will be used.

Patent Considerations:

The novel method of integrating biophysical suitability prediction with economic optimization and risk assessment for smallholder contexts may be patentable. A prior art search will be conducted before filing.

Trademark:

The system name and logo will be trademarked to establish brand identity and prevent unauthorized use.

Database Rights:

The compiled database of Sri Lankan soil-plant suitability relationships, price histories, and cultivation practices represents valuable intellectual property protected under database rights.

Confidentiality:

Farmer data remains the property of farmers, with access governed by clear terms of use and consent agreements. Anonymized aggregated data may be used for research and commercial purposes.

Open-Source Strategy:

Core machine learning models and APIs may be open-sourced to encourage academic validation and adoption, while the mobile application and optimization algorithms remain proprietary.

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